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4 / 4 submitted

Level 2 — 25 marks/question

C

CODE

```
#include <stdio.h>

int main() {
    int n, target, i, j, k, l, temp;
    int arr[100];
    int printed = 0;

    if (scanf("%d", &n) != 1) return 0;

    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &target);

    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }

    printf("Quadruplets: ");

    for (i = 0; i < n - 3; i++) {
        if (i > 0 && arr[i] == arr[i - 1]) continue;

        for (j = i + 1; j < n - 2; j++) {
            if (j > i + 1 && arr[j] == arr[j - 1]) continue;

            for (k = j + 1; k < n - 1; k++) {
                if (k > j + 1 && arr[k] == arr[k - 1]) continue;

                for (l = k + 1; l < n; l++) {
                    if (l > k + 1 && arr[l] == arr[l - 1]) continue;

                    if (arr[i] + arr[j] + arr[k] + arr[l] == target) {
                        if (printed > 0) printf(", ");
                        printf("[%d, %d, %d, %d]", arr[i], arr[j], arr[k], arr[l]);
                        printed++;
                    }
                }
            }
        }
    }
}
```

```
        }  
    }  
}  
  
if (printed == 0) {  
    printf("None");  
}  
printf("\n");  
  
return 0;  
}
```

INPUT

```
8  
1 0 -1 0 -2 2 2 -1  
0
```

OUTPUT

Quadruplets: [-2, -1, 1, 2], [-2, 0, 0, 2], [-1, -1, 0, 2], [-1, 0, 0, 1]

c

CODE

```
#include <stdio.h>

struct Student {
    char name[50];
    int roll_number;
    float marks;
};

int main() {
    struct Student s;

    scanf("%99s", s.name);
    scanf("%d", &s.roll_number);
    scanf("%f", &s.marks);

    printf("Student Details:\n");
    printf("Name: %s\n", s.name);
    printf("Roll Number: %d\n", s.roll_number);
    printf("Marks: %.f\n", s.marks);

    return 0;
}
```

INPUT

```
John
101
85
```

OUTPUT

```
Student Details:
Name: John
Roll Number: 101
Marks: 85
```

L2-Q3. substring (M)

C

CODE

```
#include<stdio.h>
#include<string.h>

int main(){
    char a[100], b[100];
    int count = 0, i, j, len_a, len_b, match;

    scanf("%99s", a);
    scanf("%99s", b);

    len_a = strlen(a);
    len_b = strlen(b);

    for(i = 0; i ≤ len_a - len_b; i++){
        match = 1;
        for(j = 0; j < len_b; j++){
            if(a[i+j] ≠ b[j]){
                match = 0;
                break;
            }
        }
        if(match == 1){
            count++;
        }
    }

    printf("Occurrences of '%s': %d", b, count);

    return 0;
}
```

INPUT

Banana
ana

OUTPUT

Occurrences of 'ana': 2

L2-Q4. triangular matrix (M)**C****CODE**

```
#include <stdio.h>

int main() {
    int rows, cols, i, j;

    scanf("%d", &rows);
    scanf("%d", &cols);

    int matrix[rows][cols];
    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    for (i = 0; i < rows; i++) {
        for (j = 0; j < cols; j++) {
            if (j > i) {
                printf("0 ");
            } else {
                printf("%d ", matrix[i][j]);
            }
        }
        printf("\n");
    }

    return 0;
}
```

INPUT

```
3
3
1 2 3
4 5 6
7 8 9
```

OUTPUT

```
1 0 0
4 5 0
7 8 9
```